MD JUBIER AHAMMED

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EDUCATION

M.S. in Geography and Environmental Studies, Texas Tech University Aug 2025 – Expected May 2027
Lubbock, TX | Current GPA: 3.863 (as of Summer 2026)
Thesis: Transferability of a Google Earth Engine Land Cover Classification Framework Between the Ganges and Mississippi Deltas.

B.S.S. in Disaster and Human Security Management, Bangladesh University of Professionals Jan 2020 – Jan 2024
Dhaka, Bangladesh | GPA: 3.90/4.00
Undergraduate thesis: Assessing and Prediction of Spatio-temporal Changes of LULC using CA-ANN Model and its Socioeconomic Implications for Coastal Bangladesh: A Case Study of Dacope Sub-district

RESEARCH INTERESTS

Natural hazards and disaster risk; remote sensing and GIS; machine learning and GeoAI; spatial analysis; social vulnerability and climate adaptation; land use / land cover change; coastal wetland mapping.

PUBLICATIONS

Peer-Reviewed Journal Articles

Ahammed, M.J., Xu, C.*, Xu, Y., Zhou, H. What Daily Nighttime Lights Can and Cannot Show about Power Outage and Recovery: Winter Storm Uri in Harris County, Texas. ISPRS International Journal of Geo-Information, 15(11), 491 (2026). https://doi.org/10.3390/ijgi15110491

Under Review

Ahammed, M.J.*, Sonet, M.S. Social Vulnerability and Compound Heat-Flood Exposure in Houston, Miami, and Norfolk: A Census-Tract-Level Analysis. Under review, Regional Environmental Change. Preprint: https://doi.org/10.21203/rs.3.rs-10349570/v1

**Corresponding author*

In Preparation

Xu, C., Ahammed, M.J. Tracking Coastal Wetland Dynamics in the Ganges Delta from 1985–2026 Using Google Earth Engine. In preparation.

Conference Presentations

Spatial Pattern of Post-storm Recovery Revealed by Nighttime Light: Evidence from Harris County, Texas. Poster presentation, Southwest Association of American Geographers (SWAAG) Annual Meeting, 2026.

Dynamic Trends in Land Use and Land Cover Change with QADI Validation: Exploring the Shifts in Shrimp Farming and Socioeconomic Implications in Coastal Bangladesh's Dacope Sub-district. International Conference on Environmental Science and Resource Management (ICESRM). Presented; no online proceedings available.

RESEARCH EXPERIENCE

Graduate Research, Department of Geosciences, Texas Tech University Aug 2025 – Present
Supervisor: Dr. Chao Xu

- Conducted a spatial and quantile analysis of infrastructure precarity and social vulnerability during Winter Storm Uri in Harris County, using VIIRS Black Marble nighttime light deficit data against ACS and social vulnerability variables; methods included Spearman correlation, OLS with clustered standard errors, robust regression, quantile regression, Moran's I, and a spatial lag model.
- Developing a Landsat random forest classification of tidal flats and wetlands in the Ganges Delta in Google Earth Engine, using manually collected ground-truth points and a revised land-cover scheme.

Independent Research Collaboration 2025 – Present
With M. Shahriar Sonet (Ph.D. candidate, Geospatial Information Sciences, University of Texas at Dallas)

- Conducting a census-tract-level analysis of compound heat-flood exposure and social vulnerability across Houston, Miami, and Norfolk, integrating NFIP flood claims, CDC social vulnerability data, NHGIS tract crosswalks, and GEE-derived land surface temperature.

Research Consultant, Remote Sensing Division, CEGIS Jan 2024 – Dec 2024

Center for Environmental and Geographic Information Services, Dhaka, Bangladesh

- Created, updated, and digitized land use / land cover maps for Chittagong division.
- Georeferenced satellite imagery for digitization.
- Managed databases and resolved topological errors.
- Conducted crop pattern analysis.

Research Assistant, Geospatial Health Studies Division, CHIRAL Apr 2023 – Dec 2023

Center for Health Innovation, Research, Action and Learning – Bangladesh

- Collected spatial and non-spatial data and conducted thematic analysis on environmental factors influencing dengue outbreaks, using GIS for disease mapping and risk assessment.
- Trained interns on mobile data collection and GIS tools (KoboToolbox, ArcGIS).
- Analyzed and visualized data in Excel and SPSS, producing charts and infographics for project reporting.

CLASS PROJECTS

Site Suitability Analysis for a Bison Preserve ("Buffalo Commons"), GIST 5300

Multi-criteria weighted suitability model in ArcGIS Pro across ten Great Plains states.

Spatial Database for EV Charging Deserts in Texas, GIST 5304

PostgreSQL/PostGIS spatial database and spatial SQL queries relating to charger locations, highways, and census tracts.

Hedonic Rent Pricing Model, Lubbock, Texas, GEOG 5309

OLS regression in R modeling monthly rent against unit attributes and neighborhood income, with diagnostics.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Department of Geosciences, Texas Tech University Aug 2025 – Present

Courses: Web Mapping and Geovisualization; Regional Geography of the World; Global Environmental Science

- Grade coursework and exams.
- Assist students during lab sessions.
- Hold weekly office hours.

AWARDS, FELLOWSHIPS & FUNDING

Second Place, Graduate Poster Competition, SWAAG Annual Meeting (2026)
Graduate Teaching Assistantship, Texas Tech University (Aug 2025 – Present)
Department of Geosciences Scholarship, Texas Tech University (Fall 2026)
Graduate Field Experience Fellowship, Texas Tech University (Summer 2026)
Gary Elbow Scholarship, Department of Geosciences, Texas Tech University (Fall 2025)

TECHNICAL SKILLS

GIS & Remote Sensing: ArcGIS Pro, ArcMap, QGIS, Google Earth Engine, ENVI
Programming & Databases: Python, JavaScript (Google Earth Engine), R, SQL/PostGIS
Statistics: SPSS, Jamovi
Remote Sensing Data: Landsat, Sentinel, MODIS, VIIRS Black Marble
Data Collection: KoboToolbox
Graphics: Adobe Illustrator, Canva

LANGUAGES

Bengali — native
English — fluent (IELTS overall band 7.5)